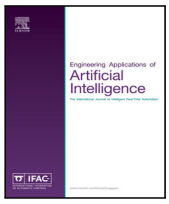
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Research paper



Information extraction from multi-layout invoice images using FATURA dataset

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ABSTRACT

Document analysis and understanding models often require extensive annotated data to be trained. However, various document-related tasks extend beyond mere text transcription, requiring both textual content and precise bounding-box annotations to identify different document elements. Collecting such data becomes particularly challenging, especially in the context of invoices, where privacy concerns add an additional layer of complexity. Invoices are critical documents in many business processes, but existing datasets for invoice analysis are limited in size and diversity, hindering the development of robust models. Current datasets do not adequately address the need for diverse layouts and comprehensive annotations, which are essential for training models capable of handling real-world variations in invoice documents. In this paper, we introduce FATURA, a pivotal resource for researchers in the field of document analysis and understanding. FATURA is a highly diverse dataset featuring multi-layout, annotated invoice document images. Comprising 10,000 invoices with 50 distinct layouts, it represents the largest openly accessible image dataset of invoice documents known to date. We provide an extensive evaluation using different information extraction methods under diverse training and evaluation scenarios, including precision, recall, and F1-score. The evaluation includes a visual-based approach using object detection for text region classification, a multi-modal strategy integrating visual and textual data for granular content comprehension, and a hybrid approach combining these methods. The dataset is freely accessible at this https://zenodo.org/record/8261508, empowering researchers to advance the field of document analysis and understanding.

1. Introduction

In an age dominated by digital information, document analysis and understanding are fundamental pillars of information retrieval and knowledge extraction (Cheikhrouhou et al., 2021). The ability to automatically process, interpret, and extract insights from documents has far-reaching applications across various domains, including finance, healthcare, legal, and administrative sectors. Invoices are one of the most prevalent and important types of documents in this context, as they represent a significant portion of the financial transactions that occur between businesses.

Invoices contain a wealth of information that can provide valuable information about business operations, including the names of the parties involved, product descriptions, and financial information. This information is essential for tracking sales, managing inventory, and monitoring cash flow. However, analyzing and understanding invoices can be challenging due to the variability and complexity of these documents. For example, invoices from different industries come in a variety of formats and layouts, often containing sensitive data. The task of collecting and annotating a diverse set of invoice documents is not only labor intensive, but is also complicated by privacy concerns. Balancing the need for data diversity with the imperative of safeguarding privacy adds an additional layer of complexity to this endeavor.

Document analysis and understanding models are heavily relying on annotated datasets for effective training. Although basic text transcription suffices for some tasks, many document-related challenges require a deeper level of annotation. This entails not only recognizing textual content, but also precisely delineating the boundaries of different elements within a document through bounding-box annotations. These annotations facilitate the extraction of structured information, making it possible to discern crucial components of the document, such as tables, titles, and signatures.

To address these challenges, researchers have turned to generative models, such as generative adversarial networks (GANs) (Goodfellow

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et al., 2020; Esser et al., 2021; Ramesh et al., 2022), as potential solutions to generate invoice images. However, the effectiveness of generative models in generating unstructured documents, such as invoices, is limited by the lack of a clear and consistent structure in these documents.

In contrast, deep learning architectures have demonstrated remarkable efficacy in various document-related tasks. Some of these models were originally designed to process image data broadly, but have exhibited exceptional performance when applied to structured document images. This category includes Convolutional Neural Networks (CNNs) (LeCun et al., 1995; Girshick et al., 2014), and Vision Transformer (ViT) (Dosovitskiy et al., 2020; Souibgui et al., 2023).

Conversely, there are models purposefully crafted for handling document images, often adopting a multi-modal approach. These models not only analyze the visual aspects of the image, but also dive into the textual content and even the layout information. An illustrative example is the LayoutLM architectures (Xu et al., 2020a,b; Huang et al., 2022).

However, it is crucial to note that these systems often rely on cutting-edge deep learning models. Deep learning models, renowned for their data-hungry nature, require substantial amounts of high-quality data for effective training. Consequently, the growing demand for robust systems and models underscores the need for large, well-annotated datasets. Acquiring datasets for invoice processing is a challenging and time-consuming task, which requires significant manual effort to extract and annotate relevant information from invoices.

Existing datasets for invoice recognition, such as Huang et al. (2019) and Park et al. (2019), are constrained by several limitations, particularly in terms of sample size and diversity of invoice layouts. Moreover, certain publicly available datasets may not contain the level of annotation required for specific tasks. Some datasets, for instance, may solely support document classification, lacking the essential bounding-box coordinates and textual details required for more comprehensive analysis. Furthermore, even when a high-quality dataset of document images is accessible, it might fall short in terms of quantity, particularly when targeting a specific document type, like invoices. This shortfall presents challenges, as certain tasks demand a robust supply of annotated document images of a particular category.

In light of these challenges, we introduce FATURA, an exemplary contribution to the realm of document analysis and understanding. FATURA constitutes a groundbreaking dataset meticulously crafted to address the prevailing limitations. This monumental dataset comprises a staggering total of 10,000 invoice document images, each adorned with one of 50 unique layouts, making it the most extensive openly accessible invoice document image dataset. This resource caters to the needs of researchers by not only offering diversity in data but also presenting an extensive benchmark for various document analysis tasks. Our aim is to provide the research community with an indispensable tool, empowering them to advance the field of document analysis and understanding while respecting the ever-important considerations of data privacy.

In the following sections, we start by providing an in-depth review of related works, where we introduce various document understanding approaches and examine the landscape of existing datasets in the field. Next, we explore the core aspects of FATURA, providing detailed insights of its content, how it is annotated, and the specific challenges it aims to overcome. Additionally, we conduct an evaluation of cutting-edge methods designed for the automatic extraction of named entities from invoices under diverse training and evaluation scenarios.

2. Related work

In the rapidly evolving field of document analysis and understanding, researchers have explored various systems, methodologies, and datasets to improve the automation of document processing tasks. This section delves into the extensive body of related work, starting with an investigation of different systems and methodologies for document understanding. We then dedicate the second subsection to a comprehensive exploration of existing invoice datasets.

2.1. Document understanding approaches

Document understanding encompasses a wide range of tasks, including layout analysis, information extraction, named entity recognition, and document image classification. Numerous systems and methodologies have been developed to address these challenges. This research field has witnessed significant advancements, driven primarily by deep learning techniques. These approaches have revolutionized the field by achieving remarkable results in a wide range of document analysis tasks. Various architectural paradigms, spanning image-based, text-based, and multi-modal models, have contributed to these successes.

Image-based architectures, exploit the success of deep CNN and object detection models (Biswas et al., 2021) to decompose the document image into semantically meaningful regions such as tables, graphical elements, titles, paragraphs... Such models excel at capturing intricate patterns, textures, and spatial relationships within visual content. For example, in Bulzan and Cernăzanu-Glăvan (2022), the authors consider the task of information extraction from invoices as an object detection task. To this end, they used three different models YOLOv5, Scaled YOLOv4 and Faster R-CNN (Girshick, 2015) to detect key field information in invoices. In addition, they propose a data preprocessing method that helps to better generalize learning. The authors in Biswas et al. (2021) focused on going beyond object detection to understand document layouts. They propose an instance segmentation model which, in contrast to classical document object detection pipelines, provides a more fine-grained instance-level segmentation for every individual object category of a document image. Image-based approaches that rely on object detection and semantic segmentation have a notable limitation; they predominantly focus on visual aspects of the document while potentially neglecting critical textual information. They may not inherently understand the arrangement of text, the significance of specific text regions, or the semantic context of the document's content. This limitation becomes especially prominent in complex documents with varied layouts and fonts.

Text-based approaches involve the conversion of a document image into a textual representation, with the choice between Optical Character Recognition (OCR) or Handwritten Text Recognition (HTR) depending on the nature of the document (printed or handwritten). Subsequently, sophisticated natural language processing techniques are applied to parse the resulting text and extract the named entity tags (Dhiaf et al., 2021). One significant weakness of this approach lies in its vulnerability to errors during the text recognition stage, particularly when dealing with low-quality scans. These errors can substantially compromise the overall performance of the subsequent Natural Language Processing (NLP) stage. In an effort to mitigate this challenge, the authors of Rouhou et al. (2022) introduced a transformer-based model to jointly perform text transcription and named entity recognition from the document images without an intermediate HTR stage. This innovative approach begins with using a CNN to extract visual features from the input images. Subsequently, these features are fed into a transformer encoder to generate hidden representations. Finally, a decoder is employed to transform these representations into a sequence of transcribed characters and named entity tags.

Recent advances in document understanding have witnessed the emergence of end-to-end models that seamlessly integrate image and text data, offering a holistic approach to document understanding within multimodal architectures. For example, the LayoutLM variants (Xu et al., 2020a,b; Huang et al., 2022), UDoc (Gu et al., 2023), and UDOP (Tang et al., 2023), go beyond mere visual processing. Take, for instance, LayoutLM in its versions 2 (Xu et al., 2020b) and 3 (Huang et al., 2022), which incorporates a spatial-aware self-attention mechanism into the Transformer architecture. This enhancement allows the model to gain a deeper understanding of the relative positional relationships among various text blocks. These models, often pre-trained in a self-supervised manner, exhibit remarkable performance in a spectrum

of downstream tasks. These tasks include form understanding, receipt comprehension, document visual question answering, document image classification, and document layout analysis.

Similarly, the groundbreaking DONUT (DOcumeNt Understanding Transformer) model (Kim and Hong, 2022), introduced a single-stage approach to document understanding. DONUT encompasses the analysis of document layouts to detect writing areas, text recognition using a lexicon of subwords, and named entity detection using specific TAGs, bolstered by a powerful external language model (BART). DONUT is pre-trained on synthetic documents, with ground truth provided as a sequence of subwords and TAGs, eliminating the need for segmentation ground truth. This paradigm shift simplifies Document Understanding to a task of learning a tagged language, assuming that the system possesses vision capabilities to construct high-level visual representations. In a similar vein, Adobe introduced DESSURT (Davis and Morse, 2022), following a comparable approach to integrate OCR and entity recognition. Additionally, the Pix2struct architecture (Kenton Lee, 2023) falls into the same category of integrated systems for document understanding. These innovations mark significant strides in document analysis streamlining and underline the evolving landscape of document understanding methodologies. However, the effectiveness of an invoice processing approach still depends on the quality and diversity of available data. In the following section, we provide a list of existing invoice datasets.

2.2. Existing datasets

In this section, we shift our focus towards a comprehensive exploration of existing datasets tailored specifically for invoice document images analysis and understanding. Within this domain, researchers have access to some document image datasets, as outlined in Table 1. Nevertheless, it is vital to emphasize that only a select few of these datasets are notably well-suited for information extraction from invoices.

For example, consider the IIT-CDIP dataset (Lewis et al., 2006) and its subset, RVL-CDIP (Harley et al., 2015). These datasets provide extensive image collections, covering a diverse array of document types, including invoices. However, it is important to acknowledge that these images commonly exhibit noise and may have relatively lower resolutions. Furthermore, the data set only includes document class labels for document image classification, lacking crucial bounding-box coordinates and textual details necessary for a more comprehensive analysis.

The SROIE dataset (Huang et al., 2019) contains a set of 1000 scanned receipt images, each accompanied by its respective annotations. Some receipts exhibit complex layouts, reduced quality, low scanner resolutions, and scanning distortions, making the data set more challenging. Another receipt dataset is CORD dataset (Park et al., 2019), featuring an extensive collection of more than 11,000 fully annotated images depicting Indonesian receipts. However, it is important to acknowledge that the SROIE and CORD datasets may exhibit limited diversity in terms of layouts, as many of the receipts follow similar structural patterns. Compared to receipts, invoices present distinct challenges for information extraction due to their diverse layouts and structured data presentation. Invoices commonly feature complex elements, such as tables, as well as multiple sections encompassing billing information, shipping details, and itemized lists. Given these differences, the SROIE and CORD datasets may not be the most suitable choice for analyzing and understanding invoice document images.

Furthermore, the FUNSD dataset (Jaume and Ekenel, 2019) offers a collection of 199 images in scanned form that feature diverse layouts and varying levels of noise. Remarkably, the dataset boasts extensive and accurate annotations, furnishing valuable ground-truth information for form understanding in noisy scanned documents. However, it is essential to recognize that the dataset's relatively modest size poses a notable limitation, which may be challenging for certain machine learning models that require a larger and more diverse dataset to be trained. Furthermore, it is essential to note that the forms in the FUNSD dataset span diverse fields, such as marketing, advertising, and scientific reports, introducing a distinct layout variety that distinguishes them from typical invoice formats.

Table 1
Summary of existing invoice-related datasets.

	Detailed annotations	Large size	Diverse	Contains invoices
IIT-CDIP	No	Yes (7 m)	Yes	Yes
RVL-CDIP	No (document-level labels)	Yes (400k)	Yes	Yes
SROIE	Yes	No (1k)	No	No (receipts)
FUNSD	Yes	No (199)	Yes	Yes
CORD	Yes	Yes (+11k)	No	No (receipts)

Existing invoice datasets, while valuable for model training and evaluation, often suffer from significant limitations. These issues primarily revolve around limited diversity, with datasets frequently skewed towards specific domains or regions. Such biases can lead to reduced accuracy when applying models to invoices from various contexts. Furthermore, an imbalanced data distribution is a common challenge, where some data sets contain a disproportionate number of a certain type of invoices, affecting model performance. Additionally, the lack of variability in invoice formats within datasets can hinder a model's adaptability to different layouts and structures. These challenges underscore the pressing need for more comprehensive and inclusive datasets to enhance the robustness and effectiveness of invoice processing models.

2.3. Challenge with the existing datasets

Confidentiality is a major issue with most custom datasets, as they often contain sensitive information about individuals, administrations, or business enterprises. For example, invoices, which include buyer names, GST numbers, and other confidential fields, are private documents for any organization. This makes the availability of such datasets challenging (Eberendu et al., 2016). Additionally, there are limited publicly available datasets, such as those provided by workshops and conferences like ICDAR, which require separate registration for access (Adnan and Akbar, 2019b; Zhao et al., 2019). The quality of datasets is also crucial, as machine learning model accuracy heavily relies on the quality of training data. Many existing datasets suffer from issues such as distorted, skewed, blurred images, and noisy backgrounds, leading to poor OCR performance (Palm et al., 2019; Adnan and Akbar, 2019a). Furthermore, outdated datasets, like RVL-CDIP, with obsolete document structures and formats, blurry images, and missing data values, pose significant challenges for information extraction and affect model performance. Moreover, Existing datasets primarily cater to specific domains such as Scientific Literature or the Medical domain, focusing on task-specific metadata extraction tasks like Article Title or Bibliography Extraction, or Patient Details Extraction. Some datasets include images with handwritten characters, presenting challenges due to the variability in handwriting styles, which complicates text recognition and requires advanced OCR techniques (Davis et al., 2019). Additionally, language-specific datasets, such as French receipts, Chinese passports, identity cards, and patient medical receipt datasets, present challenges due to ambiguous language and incorrect morphological phrases (Ye et al., 2018; Alfred et al., 2014). Another significant issue is the absence of labeling or annotations, with many public datasets lacking a large, labeled corpus, making domain-specific annotations a critical challenge (Adnan and Akbar, 2019b).

To tackle these challenges, we introduce FATURA, a multi-template invoice document images dataset. This dataset exhibits a remarkable

diversity in invoice layouts, aiming to significantly enhance the capabilities of models for analyzing and understanding unstructured documents. Our aim is to enhance the performance of these models, ensuring their effectiveness in real-world applications.

3. FATURA dataset

In this section, we introduce the FATURA dataset, illustrated in Fig. 2. We describe the generation process of the documents, their structure and their content. The database containing both the images and the annotations is freely available at https://zenodo.org/record/8261508.

3.1. Invoice generation process

The generation of synthetic invoice templates is a complex yet crucial step in creating a diverse and representative dataset for document analysis and understanding. This process involves multiple stages, each carefully designed to strike a balance between realism and diversity while addressing crucial privacy concerns. The generation process revolves around the transformation of a blank canvas into plausible invoice templates. This intricate procedure consists of several key steps aimed at producing diverse and representative templates.

The first step is to gather a comprehensive dataset of real invoice images. These genuine invoice templates serve as the fundamental building blocks for our synthetic templates. The images are of varying templates that differ in terms of both layout and content. To ensure that the generated templates encapsulate essential components, such as buyer information, total amounts, and invoice dates, the templates are carefully selected based on their content.

Secondly, each real image is annotated using the VGG Image Annotator tool. The primary objective of this annotation process is to establish a well-annotated layout for each invoice template. At this stage, the textual content is omitted because it will be generated randomly in subsequent phases. Our focus is on capturing the structural blueprint of the invoices, including the placement and dimensions of different elements. We also incorporate logos into our synthetic templates to enhance their realism. A unique logo is generated for each template using a pre-trained open-source Text-to-Image Latent Diffusion model (Rombach et al., 2022). It is important to note that all images from the same template share a common background color, adding to the visual coherence of the dataset.

Generating variants of each invoice template is a key step in enriching the dataset. The algorithm begins with a blank canvas and proceeds to insert the relevant textual information in each component of the original template. For each component, a bounding box is created according to the original coordinates, preserving the spatial arrangement of the elements. The text generation process is carefully tailored to create invoices that mimic real-world diversity. To further enrich the dataset's realism and variability, certain components such as sender/receiver names, addresses, and product descriptions are populated with randomly selected text from a predefined repository of plausible texts. The algorithm also verify that the total amount on the invoice matches the sum of all amounts of individual products.

3.2. Dataset description

The dataset consists of 10000 JPEG images each accompanied by their corresponding JSON annotation files. These images are generated based on a set of 50 distinct templates. The selection of the 50 distinct layouts was based on several key criteria to ensure maximum diversity in the dataset. First, we prioritized a wide range of invoice formats from different industries, incorporating variations in design, structure, and content. This included differences in header and footer placements, logo positions, table formats, font styles, and the arrangement of key text regions such as addresses, dates, and totals. This diversity reflects real-world scenarios and ensures the relevance of the data set to a wide range of applications. It is noteworthy that even within the same template, each generated invoice image exhibits distinct textual content, further enhancing the dataset's realism and utility.

We have identified 24 distinct classes corresponding to the different fields that can be extracted from an invoice, as indicated in Table 2. It should be noted that the frequency of these classes exhibits considerable variation, leading to an imbalanced dataset, as visually depicted in Fig. 3.

All invoices in the dataset are in English, and use either the dollar or the euro as currency. While the invoices vary significantly in their layouts, they are not categorized by specific types or classes of invoices. Instead, they are organized based on distinct templates, reflecting a range of structural differences rather than predefined invoice categories.

As shown in Fig. 1, the templates exhibit a wide range of layouts. Equivalent text regions are positioned differently across templates, with variations in table structures, background colors, and logo placements. Additionally, not all templates contain the same content, meaning certain text regions appear in some templates but are absent in others.

For example, as in Fig. 1, templates 15 (top left) and 47 (top right) differ in several aspects. The logos are in different positions (one is at the bottom of image and one is in the top right corner) and the tables are of different sizes. Furthermore, the dates in template 15 are in the top right region, but are on the left and just above the table in template 47.

Templates 15 (bottom left) and 45 (bottom right) are also notably different in terms of layout. The table in template 15 features text that is widely spaced and a bold font for column titles, whereas the one in 47 is more compact with a colored top row.

Annotations for the FATURA dataset are available in three different formats to accommodate various research needs. The first format adheres to the well-established COCO annotation format, widely recognized and utilized in computer vision tasks. The second format is meticulously tailored for seamless integration with the HuggingFace[1] Transformers library (Wolf et al., 2019), a renowned resource in the field. This format is specifically designed to align with the capabilities of the LayoutLMv3 (Huang et al., 2022) architecture. The third format represents our dataset's standard annotation format, which can be used in research contexts where a custom or unique annotation schema is preferred.

3.3. Evaluation strategies

In this section, we present two distinct evaluation strategies that facilitate the training and assessment of models on our dataset, each offering unique insights into model performance.

In the first evaluation strategy, we employ an intra-template-centric scenario. For each of the 50 distinct templates, the generated images are randomly partitioned into three subsets: training, validation, and testing. In this scenario, models are trained on all templates, thereby ensuring exposure to various layouts and styles. Consequently, during testing, models encounter new images based on familiar templates. This approach provides a valuable assessment of a model's ability to generalize its understanding of document layouts and adapt to diverse content.

The second evaluation strategy adopts an inter-template centric perspective, emphasizing the diversity of templates and layouts. Here, the templates are randomly segregated into a training set, and a distinct set of templates is reserved for validation and testing. This evaluation scenario evaluate the models' performance on different unseen templates/layouts, rather than the same templates with different content.

[1] https://huggingface.co.

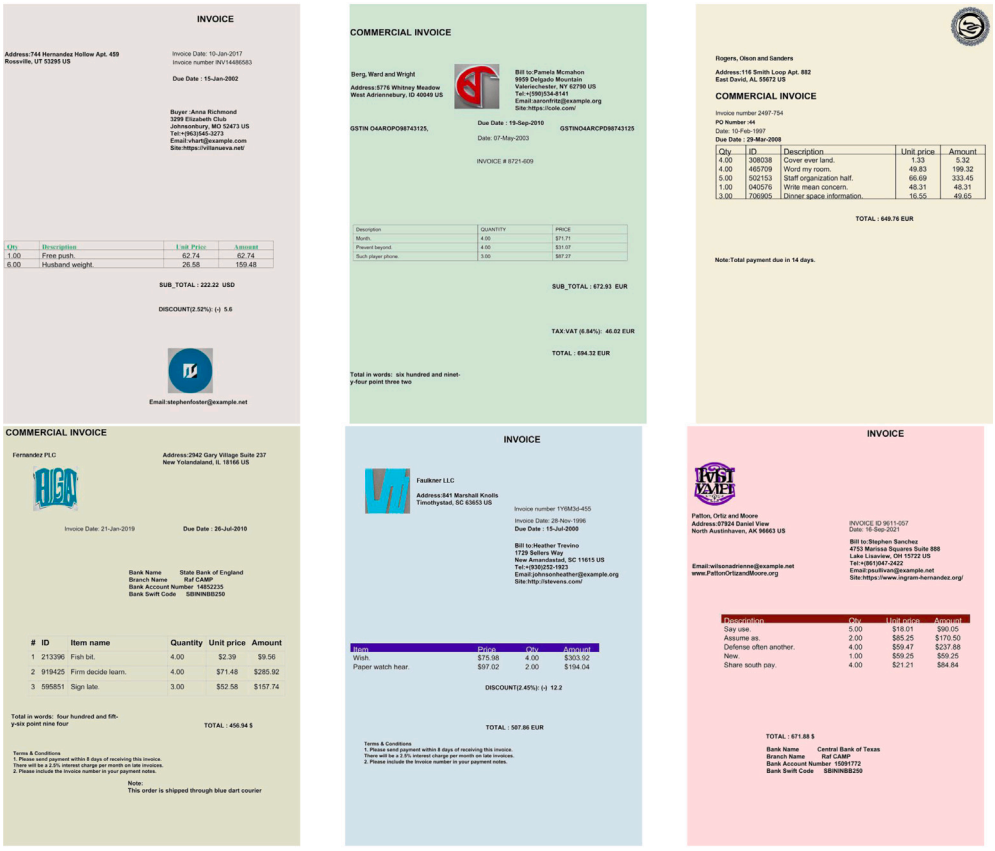


Fig. 1. Examples of images from different templates.

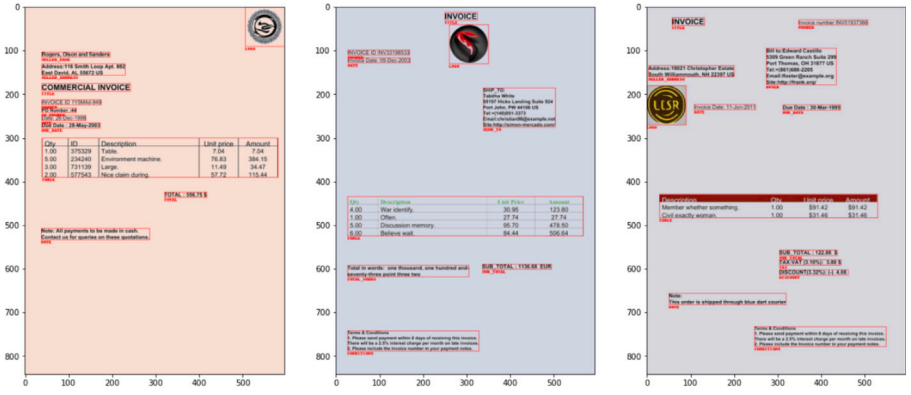


Fig. 2. Examples of annotated images from different templates.

By embracing these two complementary evaluation strategies, we ensure a comprehensive assessment of the performance of the model. The first strategy focuses on the adaptability of models to diverse content within familiar templates, while the second strategy challenges models to generalize effectively across a variety of templates and layouts, evaluating their adaptability to new, unseen document structures. This approach facilitates a holistic evaluation of models in the context of real-world document analysis and understanding tasks.

4. Experimental results

In this section, our objective is to establish comprehensive benchmarks by training and testing models on a spectrum of document layout analysis and document understanding tasks. We aim for future research using this dataset to leverage these results as benchmarks for comparison. The following section will introduce the various models evaluated in this study, detailing their architectures and performance on the aforementioned tasks.

4.1. Models description

In this section, we present the different models evaluated on the proposed dataset, covering a range of architectures designed for document layout analysis and document understanding tasks. We will explore both traditional and state-of-the-art models, including object detection, multi-modal, and hybrid approaches, providing a detailed breakdown of their design, training processes, and the performance metrics used to assess their effectiveness.

- Vision-based Object Detection approach: In this approach, we leverage object detection techniques to classify text regions within

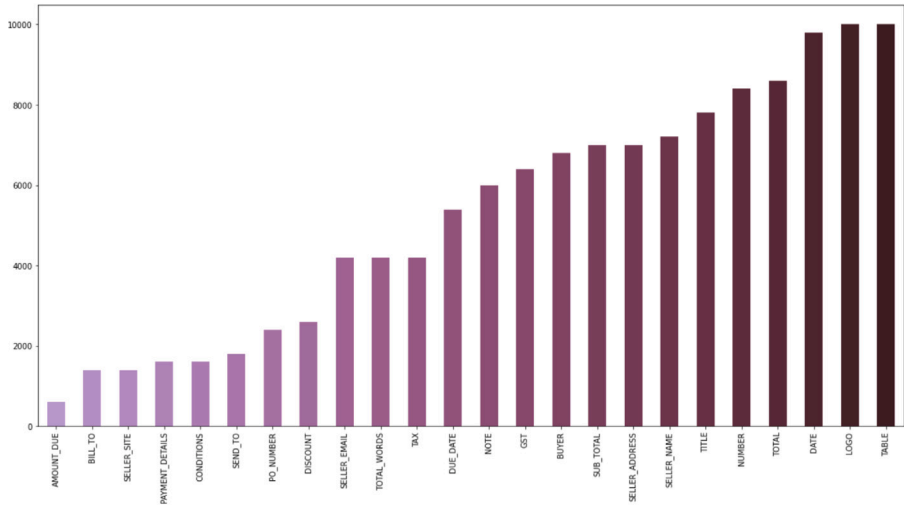


Fig. 3. Class occurrence distribution in the FATURA Dataset.

Table 2
Description of the information contained in the different invoices.

Class/text-region	n°Instances	% among objects	n°Templates	Description
TABLE	10 000	8%	50	Information about bought products
LOGO	10 000	8%	50	Template logo
DATE	9800	7.9%	49	Date of purchase
NUMBER	8800	7%	44	Invoice ID
SELLER ADDRESS	8200	6.6%	41	
TOTAL	8000	6.4%	40	Total amount after tax and discount
TITLE	7400	5.9%	37	
SELLER NAME	6800	5.5%	34	
SUB-TOTAL	6800	5.5%	34	Total amount before tax and discount
BUYER	6200	5%	31	Buyer information
DUE DATE	5800	4.7%	29	
NOTE	5200	4.2%	26	Remarks and footers
GST	5000	4%	16	Goods and services tax
SELLER EMAIL	4400	3.5%	22	
TOTAL WORDS	4000	3.2%	20	Total in words
TAX	3800	3%	19	
PAYMENT DETAILS	2600	2%	13	Bank information
DISCOUNT	2400	1.9%	12	
CONDITIONS	2400	1.9%	12	Payment terms and conditions
SEND TO	1800	1.4%	9	To whom the invoice is sent
BILL TO	1600	1.3%	8	To whom the bill is sent
PO NUMBER	1400	1.1%	7	Purchase order number
SELLER SITE	1400	1.1%	7	Website of the seller
AMOUNT DUE	800	0.6%	4	Total amount to be paid

document images. This method relies solely on visual cues and layout information to analyze and categorize document content. We use the YOLOS transformer model (Fang et al., 2021) to provide benchmarks for this task. The model is a transformer encoder model taking in image patches as inputs and producing a number of "detection tokens" as outputs. "Detection tokens" are embeddings that are passed on to two MLP heads, one for classification and one for bounding box regression. The metrics used for evaluation are standard object detection metrics: Mean Average Precision at IOU thresholds 50% and 75%, as well as Mean Average Recall at different thresholds (10 and 15) of maximum detected regions per image.

- Multi-Modal Token Classification approach: In contrast, our second approach employs a multi-modal strategy for token-level classification. Here, we integrate both visual, textual and layout information to classify and understand document content at a more granular level. We leverage the LayoutLMv3 (Huang et al., 2022) model to assign a text-region class to every text token in a given invoice. The final class assigned to a given text region is the most common class assigned to its individual text tokens. The model is a multimodal transformer encoder. Its input is a sequence of text tokens, that are embedded and aggregated with a sequence of positional encodings, and followed by a sequence of positional encoding of image patches. The positional embeddings of text tokens are of 2 kinds: embeddings representing the 2D bounding box coordinates of the text regions, and embeddings representing the 1D position/order of each token in its corresponding text region. A separator token separates the text-layout embeddings from the image patch embeddings. As this is a task of document-image token classification, we employ classification metrics for evaluation. Specifically, we use recall, precision, accuracy and F1-Score. It is worth noting that in this task, the ground

truth textual information is assumed to be known and is fed to the model alongside the ground truth image. The performance of the LayoutLMv3 Model in the first evaluation scenario is not presented in a table as model testing resulted in values of 99% in all metrics for both the validation and test sets.

- Hybrid Token Classification approach: Additionally, we explore the synergies between these approaches by combining object detection and token classification methods. The motivation for this approach is that the token classification approach assumes the ground truth text is known. As this is not always the case in real life scenarios, the textual content of the invoice is extracted using an OCR engine. Specifically, we employ the YOLOS model fine-tuned in the previous experiment to detect text regions, which are then cropped and passed to the EasyOCR (Gu et al., 2023) engine to extract text. The extracted text is passed alongside the image to the LayoutLMv3 model to classify tokens. The token-wise labels are used to produce region-wise labels as described in the previous LayoutLMv3-based task.
- OCR-Free Document Image Parsing approach: We leverage a transformer model to extract important information from invoice images. Specifically, the model is to extract the textual content of text regions that are often of high relevance in invoice information extraction tasks. The extracted information is then parsed in a JSON format. The model we used here is the transformer model Donut (Kim and Hong, 2022), which consists of a vision transformer to extract features from images, and a decoder to generate token sequences. The metrics used for this task are recall, precision, F1-score and Tree Edit Distance (TED)-based accuracy.
- Vision-based Sequence Modeling approach: In this approach, we leverage a sequence modeling approach to localize and classify text regions in a given document image. We leverage the recently published ViTLP model (Mao et al., 2024) to generate classes and bounding boxes for text regions. The model consists of a vision transformer encoder, as well as a decoder comprising a token classification head and a bounding-box regression head. The token classification head is originally intended to transcribe text tokens in a given text region, but as the task we are tackling is not an OCR task, we used tokens to represent text-region classes rather than text tokens. This means that the token embedding matrices were replaced by new ones representing embeddings for text-region classes. The metrics used for evaluation are the same object detection metrics used with the YOLOS-based object detection approach: Mean Average Precision at IOU thresholds 50% and 75%, as well as Mean Average Recall at different thresholds (10 and 15) of maximum detected regions per image.

4.2. Results and performance evaluation

In this section, we present the results of our model evaluations across various document layout analysis and document understanding tasks. We provide detailed performance metrics, including accuracy, precision, recall, and F1 scores, to assess the effectiveness of each model. These results will offer valuable insights into the strengths and limitations of the models, serving as a benchmark for future work that seeks to build upon this dataset.

4.2.1. Vision-based object detection

To adapt YOLOS for our specific task, we fine-tune the YOLOS-Ti version (the smallest version of YOLOS), pre-trained on COCO 2017 (Lin et al., 2014).

In Table 3, we present the results for the first evaluation strategy (intra-template evaluation), while Table 4 displays the results for the second evaluation strategy (inter-template evaluation). The results include mean Average Precision (mAP) scores at a specific IOU value. Additionally, we report the averaged maximum recall (mAR) which represents the maximum recall given a fixed number of detections per image, averaged over categories and IoUs.

Table 3
YOLOS results: First evaluation strategy.

Metric	Training	Validation	Test
mAP@IOU = 50	97.57%	97.77%	97.57%
mAP@IOU = 75	81.92%	81.31%	81.84%
mAR@maxDets = 10	78.05%	77.55%	77.73%
mAR@maxDets = 15	78.08%	77.56%	77.75%

Table 4
YOLOS results: Second evaluation strategy.

Metric	Training	Validation	Testing
mAP@IOU = 50	99.1%	43.6%	43.58%
mAP@IOU = 75	97%	32.66%	32.7%
mAR@maxDets = 10	90.17%	32.14%	32.14%
mAR@maxDets = 15	90.2%	32.14%	32.15%

The results obtained from our experimentation reveal a notable trend: the first evaluation strategy, which emphasizes intra-template evaluation, consistently outperforms the second evaluation strategy focused on inter-template assessment. The central factor that contributes to the observed performance disparity lies in the inherent capabilities of the YOLOS model. In fact, YOLOS, by design, excels at analyzing the layout and structure of document images and succeeds in identifying and categorizing complete text regions within the documents based on visual cues, rather than comprehending textual content. In the first evaluation strategy, where the model encounters new images based on the same templates it has been trained on, the model showcases its strength in intra-template understanding. It has learned the intricate layouts and text region placements specific to each template during training. Consequently, when tasked with analyzing new instances of familiar templates during testing, the model excels in accurately localizing and classifying text regions. On the contrary, the second evaluation strategy challenges the model with templates that it has not seen during training. Although the model has a strong foundation in layout analysis, it faces difficulties when faced with entirely new templates that deviate from the ones it has been exposed to. This inter-template generalization proves to be a more demanding task, as it requires the model to adapt to diverse layouts, structures, and visual cues that may not align with its training data.

These results have significant implications for the application of detection-based models in document analysis. Models like YOLOS, designed for layout analysis and object detection, excel at tasks related to visual recognition and structuring, but do not possess text-completion capabilities. Therefore, their performance is highly dependent on the familiarity of the templates. In conclusion, these results emphasize the importance of adopting a nuanced approach that incorporates advanced text comprehension capabilities, such as those offered by LayoutLMv3.

4.2.2. Multi-modal token classification

We harness the power of the LayoutLMv3 architecture to create a multi-modal framework capable of modeling the intricate interactions among text, layout, and image components within document images. This approach represents a significant departure from the purely visual-based strategy discussed earlier, as it embraces a holistic understanding of documents by integrating textual, layout, and visual information.

Although good performance has been achieved with LayoutLMv3, domain knowledge of one document type cannot be easily transferred to another. In addition, one significant drawback of LayoutLM is its reliance on a complex preprocessing step for word-bounding-box segmentation. Document images like invoices featuring a complex and varied layout (key–value pairs in a left–right layout, tables in a grid layout, etc.) often need to undergo extensive preprocessing to identify and delineate individual word bounding boxes accurately. This step can

Table 5
LayoutLMv3 results: Second evaluation strategy.

Split	F1-score	Recall	Precision
Training	100%	100%	100%
Validation	95.7%	97.5%	95.7%
Testing	95.7%	97.5%	95.7%

be computationally expensive and may require additional expertise in data preparation, making the model less accessible for users without specialized knowledge.

On the other hand, LayoutLM's performance is heavily reliant on the quality of Optical Character Recognition (OCR) results. Inaccuracies or errors in text recognition by the OCR system can negatively impact the model's ability to understand document content. LayoutLMv3 assumes that OCR provides precise and complete text transcriptions, which may not always be the case, especially when dealing with complex documents such as invoices.

In our specific scenario, we attempted to train and deploy LayoutLMv3 at the word level, utilizing the FATURA dataset. Regrettably, the results fell short of expectations, with the model struggling to precisely identify and delineate individual word bounding boxes. This posed a significant challenge to the model's performance in accurately processing the dataset.

For these reasons, we decided to apply LayoutLMv3 at the region level instead of the word level for key–value extraction in the context of invoices. This can help the model capture the contextual relationships between these regions and the associated key–value pairs. In addition, invoices can have diverse layouts, with variations in the placement of key information. By analyzing regions as a whole, LayoutLMv3 can adapt to different layout styles, making it more robust in handling invoices from various sources and formats. Working at the region level can also reduce the noise introduced by irregular spacing, formatting, or variations in text sizes within key–value pairs. The model can focus on the overall structure and content of each region, improving the accuracy of key–value extraction.

Before delving into the results of our hybrid approach, it is essential to establish a baseline reference approach. In this reference approach, we utilize region-bounding box annotations and their corresponding text ground-truth transcriptions during the inference stage. However, it is important to note that this approach is not practically viable and is primarily intended for comparative purposes. Its performance represents the upper bound as it leverages ground-truth annotations, which are typically unavailable in real-world scenarios. The purpose of introducing this reference approach is to assess and contextualize the performance of our hybrid approach, which operates under more realistic conditions. To evaluate the model performance, we report the precision, recall, and F1-score metrics.

In the context of the first evaluation strategy (intra-template evaluation), the model shows high performance, achieving remarkable 99% across all metrics for both the validation and testing sets. However, when transitioning to the second evaluation strategy, the model performance exhibits a slight decrease, as indicated in Table 5. This expected decline in performance can be attributed to the increased complexity of the second scenario, which involves diverse templates and layouts present across distinct sets.

4.2.3. Hybrid token classification

This approach combines the strengths of YOLOS object detection and the multimodal LayoutLMv3 model to achieve precise extraction of key–value information within text regions.

In the initial phase, we employ the pre-trained YOLOS model from our earlier experiment to identify text regions. Although YOLOS, like other object detection architectures, allows for overlapping bounding boxes, our generated dataset features non-overlapping text regions.

Table 6
Hybrid approach results: First evaluation strategy.

Split	F1-score	Recall	Precision
Training	88%	85%	92%
Validation	88%	85%	92%
Testing	88%	84%	92%

Table 7
Hybrid approach results: Second evaluation strategy.

Split	F1-score	Recall	Precision
Training	79.3%	70.8%	91.1%
Validation	56.3%	47.7%	70.7%
Testing	56.3%	47.7%	70.7%

To address this, we implement a modified Non-Maximum Suppression technique with an IOU threshold of 30% to eliminate redundant bounding boxes.

Subsequently, an OCR system, specifically EasyOCR (Gu et al., 2023), is used to extract text content from the cropped regions. The resulting bounding boxes and their corresponding text content are then input into the LayoutLMv3 model for token classification. The model assigns labels to individual tokens, and the most frequently occurring label (mode) is assigned as the label for the entire text region. This comprehensive approach ensures accurate and context-aware classification within the identified text regions.

This approach consists of using YOLOS object detection to detect text regions, followed by the multi-modal LayoutLMv3 model that classifies individual tokens, rather than text regions. We use the pre-trained YOLOS model from the previous experiment.

First, the YOLOS model is used to localize text-regions. The YOLOS architecture, just like other object detection architectures, allows for overlapping boxes. As the text regions in our generated dataset do not overlap, we deal with this issue by removing certain bounding boxes using a variant of Non-Maximum Suppression with an IOU threshold of 30%.

Then, an OCR system (we used EasyOCR (Gu et al., 2023)) is used to extract text from the cropped regions. The bounding boxes and the corresponding texts are then fed to the LayoutLMv3 model to get token classification labels. Finally, the most frequent label is used as the label of the entire text-region.

Similar to the preceding approaches, our evaluation of the hybrid approach, as depicted in Table 6, reveals a notable performance advantage in the context of the first evaluation scenario compared to the outcomes in the second evaluation strategy, outlined in Table 7.

When contrasting these results with those of the previous section, which employed region bounding box annotations and corresponding text ground truth transcriptions during inference, we observe a discernible performance gap. This variance can be attributed to inherent segmentation errors in the YOLOS model, coupled with OCR inaccuracies, both of which exert a detrimental impact on the model's capacity to accurately comprehend the document content. These errors introduce challenges that affect the model's ability to precisely classify tokens within text regions, particularly in the more complex and diverse context of the second evaluation strategy.

4.2.4. OCR-free document image parsing

In this task we finetune a Donut model (Kim and Hong, 2022) to extract information from a document in the form of a string that is convertible to a JSON format. The task is identical to the one showcased in the original Donut paper (Kim and Hong, 2022) utilizing the CORD dataset (Park et al., 2019). In fact, we use an instance of the model that is previously pre-trained on the CORD dataset.

The results presented in the Tables 8 and 9 demonstrate a clear trend: similar to the previously tested models, the Donut model exhibits

Table 8
Donut results: First evaluation strategy.

Split	F1-score	Recall	Precision	TED Acc.
Training	88.3%	87.6%	89.0%	97.4%
Validation	87.9%	87.1%	88.8%	97.0%
Testing	88.2%	87.5%	88.9%	97.4%

Table 9
Donut results: Second evaluation strategy.

Split	F1-score	Recall	Precision	TED Acc.
Training	85.3%	84.2%	86.4%	96.3%
Validation	75.7%	73.8%	77.7%	82.2%
Testing	73.6%	69.3%	78.4%	77.1%

Table 10
ViTLP results: First evaluation strategy.

Metric	Training	Validation	Test
mAP@IOU = 50	99.8%	99.8%	99.8%
mAP@IOU = 75	99.5%	99.6%	99.6%
mAR@maxDets = 10	95.7%	95.5%	95.7%
mAR@maxDets = 15	95.8%	95.5%	95.7%

significantly better performance in the intra-template evaluation scenario compared to the inter-template scenario. Specifically, the model shows a strong capability to extract information from invoice templates it has encountered during training, but struggles when applied to invoices with unseen layouts. This performance disparity suggests that the model has a tendency to overfit to the specific structural patterns of the familiar templates, limiting its generalizability to new, unseen formats. This overfitting behavior highlights the diversity present in our proposed dataset, both in terms of document layouts and content. The variety in template structures within the dataset likely contributes to the model's challenges in adapting to novel templates, further emphasizing the importance of generalization in document information extraction tasks.

4.2.5. Vision-based sequence modeling

In this approach, we employ a sequence modeling technique to localize and classify text regions within a given document image. Specifically, we utilize the ViTLP model (Mao et al., 2024) to generate both class labels and bounding boxes for the identified text regions.

The ViTLP model exhibits notably peculiar behavior, particularly in its contrasting performance across the two evaluation scenarios. As indicated in Table 10, the model achieves exceptional performance in the intra-template evaluation scenario, where it processes invoices with layouts it encountered during training. However, in the inter-template scenario, where the model is tested on unseen templates, its performance drops drastically, achieving results as low as 1% to 6% across all evaluation metrics. This stands in stark contrast to the model's training performance, which exceeds 90%, indicating severe overfitting to the training data.

While overfitting was observed in previous models, the ViTLP model demonstrates a much more extreme form of this behavior, suggesting that its complexity may be disproportionate to the task at hand. This extreme overfitting raises important questions about the suitability of such a complex model for this specific task, and further investigation is required to understand the underlying causes of this performance disparity.

Nevertheless, the stark gap in performance between the two evaluation scenarios serves to reinforce the considerable diversity present in our dataset, both in terms of layout and content. If the training and testing invoice images were too similar, such a significant drop in performance during the inter-template evaluation would not be observed. This highlights the challenge of generalization in document information extraction tasks and emphasizes the need for models that can handle the inherent variability in real-world invoice data.

Table 11
Average Inference Time Per Sample on an RTX3090 Nvidia GPU.

Model	Average inference time per sample
Donut	131.7 ms
ViTLP	260 ms
YOLOS	13.8 ms
LayoutLMv3	7.5 ms
Hybrid	530 ms

4.3. Computational efficiency

As computational efficiency is an important evaluation criterion for machine learning models, we have provided measures of the average inference time per sample on an Nvidia RTX3090 GPU for all models in Table 11. To calculate inference time, each model was used to perform inference on a set of images containing 1400 samples and inference time for each sample was recorded. The average inference time reported in Table 11 is the average across the entire set.

As illustrated in Table 11, models that produce outputs in a single forward pass, such as YOLOS and LayoutLMv3, are the most time-efficient, requiring the least amount of time to generate results. In contrast, models that generate outputs sequentially and in an autoregressive fashion, such as Donut and ViTLP, are comparatively less efficient in terms of inference time. Additionally, it is important to highlight that the hybrid approach is the slowest among all the methods tested. This is expected, as the hybrid model, despite being based on the lighter models, involves multiple processing steps. These include converting the outputs from one model format to another, as well as performing an additional OCR step, all of which contribute to the increased computational overhead and slower performance.

5. Limitations

While the synthetic invoice dataset provides a valuable resource for training and evaluating machine learning models, it is important to acknowledge several limitations and biases that may impact the generalizability and robustness of models trained on this data:

Lack of Real-World Variability: The synthetic invoices were generated using predefined templates and controlled parameters. As a result, the dataset may lack the variability found in real-world invoices. For instance, genuine invoices often exhibit a wide range of formatting styles, fonts, logos, backgrounds, stamps, and handwritten annotations. The controlled nature of the synthetic data may not fully capture these variations, leading to potential overfitting of models to the specific styles present in the synthetic dataset.

Simplified Noise and Artifacts: In real-world scenarios, invoices can be subject to various types of noise, such as creases, stains, smudges, or scanning artifacts (e.g., skewed text, blur, low resolution). The dataset only contains clean images. While it is possible to simulate degradations, it is difficult to accurately reproduce the complexity and randomness of noise encountered in actual scanned or photographed invoices. Consequently, models trained on this dataset may underperform when faced with real-world noise and artifacts.

Bias Towards Synthetic Generation Rules: The rules and algorithms used to generate the synthetic invoices inherently introduce biases. For example, the distribution of invoice amounts, item descriptions, and company names may reflect the predefined rules rather than the natural distribution observed in real-world data. This could lead to models that are overly attuned to the synthetic patterns and less capable of generalizing to real-world invoices where these distributions differ.

Limited Representation of Diverse Scenarios: The synthetic dataset does not cover the full spectrum of scenarios encountered in real-world invoicing. For example, it lacks representation of invoices in languages other than English, multi-currency formats, or those containing multi-page structures with complex table layouts. This limitation could result

in models that perform well on the synthetic data but struggle with the diversity of real-world invoices.

In summary, while synthetic datasets offer a controlled and scalable means of generating training data, the limitations outlined above highlight the need for careful consideration when using such data for model development. To mitigate these limitations, it is recommended to supplement synthetic datasets with real-world data and to thoroughly test models on a wide range of real-world scenarios to ensure their robustness and generalizability.

6. Conclusion

In summary, this paper has made significant strides in advancing the domain of document analysis and understanding through innovative approaches and the introduction of a valuable resource. We presented the FATURA dataset, a diverse collection of multi-layout and annotated invoice documents, addressing the critical need for high-quality, unstructured document datasets, with a particular focus on invoices that are vital in numerous real-world applications.

Throughout our investigation, we thoroughly examined various evaluation strategies and employed state-of-the-art models such as LayoutLMv3 and YOLOS to tackle intricate document analysis tasks. We introduced our hybrid approach, harnessing the synergy between object detection and token-level classification to enhance document understanding, even in the face of segmentation and OCR inaccuracies.

In conclusion, our contributions in dataset creation, model evaluation, and the proposed hybrid approach establish a robust foundation for future research in document analysis and understanding. We believe that the FATURA dataset and the insights presented in this paper will serve as valuable resources, inspiring further innovations and advancements in the field. One promising perspective is the extension of this dataset to encompass multi-lingual invoices, catering to a broader range of document types, languages and possibly units of currency. Another valuable direction would be to expand the dataset to support tasks such as Document Visual Question Answering (DocVQA), which would involve incorporating ground truth labels for question–answer pairs, further enhancing the dataset's utility for diverse document understanding challenges.

CRediT authorship contribution statement

Mahmoud Limam: Writing – original draft, Methodology, Writing – review & editing. **Marwa Dhiaf:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Yousri Kessentini:** Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data is available at https://zenodo.org/records/8261508.

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